DOCTOR'S CONSULTATION FEE ANALYSIS

Business & Exploratory Data Analysis Report

Data-driven analysis of consultation pricing across cities and medical specializations in India

Project Snapshot	Value
Data source	Practo
Records analyzed	1,080 doctors
Cities covered	12
Medical specializations	9
Final analytical columns	9
Analysis type	Exploratory & comparative analysis
Primary tools	Python, Pandas, NumPy, Matplotlib, Seaborn

Prepared as a portfolio-ready project document

1. Executive Summary

This project analyzes doctor consultation fees across 12 Indian cities and 9 medical specializations using a 1,080-record dataset collected from Practo. The objective is to understand how consultation pricing varies across location, city tier, specialization, experience, patient ratings, and premium pricing segments.

The analysis shows substantial variation in observed consultation fees across cities and specializations. Mumbai records the highest average fee at ₹1,478.89, while Patna records the lowest at ₹515.00. Metro cities average ₹968.70 compared with ₹690.00 in Non-Metro cities, a difference of ₹278.70.

At the specialization level, Cardiologists have the highest average consultation fee (₹1,016.25), while Dentists have the lowest (₹425.40). Experience and patient rating show only weak linear associations with consultation fee, with Pearson correlations of 0.142 and -0.133 respectively.

Executive takeaway
Pricing differences are much more visible across cities and specializations than through experience or patient-rating relationships. This makes geographic and service-line segmentation the most actionable descriptive lens in this dataset.

Key Metrics at a Glance

Metric	Result
Average consultation fee	₹829.35
Median consultation fee	₹800.00
Fee range	₹0 – ₹4,000
Free consultations	27
IQR fee outliers	39
Premium threshold	₹1,000
Premium records	365 (33.8%)
Experience–fee correlation	0.142
Rating–fee correlation	-0.133

2. Business Problem

Healthcare consultation pricing can vary considerably across cities and medical specializations. Without structured analysis, it is difficult to compare pricing levels, identify higher-priced markets, or understand how consultation fees differ across medical services.

- Consultation fee structures across cities and medical specializations vary considerably, making healthcare pricing difficult to compare and interpret.
- Patients may face uncertainty when choosing healthcare providers because consultation fees can differ significantly across doctors, cities, and specializations.
- Doctor-level data from digital healthcare platforms creates an opportunity to identify meaningful patterns in consultation pricing.
- Systematic analysis can reveal regional pricing differences, specialization-wise variation, premium pricing concentration, and relative fee segments.

Business Questions

- Which cities have the highest and lowest observed consultation fees?
- How different are consultation fees between Metro and Non-Metro markets?
- Which medical specializations command higher or lower average fees?
- How strongly are experience and patient ratings associated with consultation fees?
- Where are premium-priced consultations concentrated?
- Can paid consultation fees be grouped into useful relative pricing segments?

3. Objectives

1. Analyze the overall distribution of doctor consultation fees.
2. Compare consultation fees across cities and between Metro and Non-Metro markets.
3. Compare pricing across medical specializations.
4. Examine relationships between consultation fees, doctor experience, and patient ratings.
5. Identify premium-priced cities, specializations, and city–specialization combinations.
6. Segment paid consultation fees into Budget, Mid-range, and Premium relative groups.
7. Translate observed pricing patterns into business-oriented insights without making causal claims.

4. Data Source & Methodology

4.1 Data Source

The project uses Practo as the source for doctor-level information including doctor name, specialization, experience, hospital/clinic name, city, patient recommendation/rating, source, and consultation fee.

The project documentation preserves the historical 1,080-record dataset used for the analysis. The extraction notebook also contains a controlled live-scraping demonstration using Requests and BeautifulSoup. A full live extraction is disabled by default because the current Practo website may return a different set and number of records.

4.2 Historical Dataset Construction

Stage	Records
Specialization-level consolidated dataset	1,066
Additional records	14
Project raw dataset	1,080
Final analysis-ready dataset	1,080

4.3 Extraction Workflow

8. Build a Practo search URL for a specialization–city combination.
9. Send an HTTP request using Python Requests with browser-like headers and a timeout.
10. Parse the returned HTML using BeautifulSoup.
11. Extract doctor-level fields and validate record structure.
12. Consolidate extracted records into a structured dataset for downstream cleaning.

4.4 Data Pipeline

Pipeline

Web Scraping → Data Consolidation → Data Cleaning → Analytical Preparation → Exploratory Data Analysis

5. Dataset Profile

The final analysis-ready dataset contains 1,080 records and 9 columns. It covers 12 cities, 9 medical specializations, and two city tiers.

Dimension	Coverage
Records	1,080
Cities	12
Specializations	9
City tiers	Metro / Non-Metro
Source	Practo
Duplicates	0
Missing values in analysis dataset	0

Dataset Fields

Column	Business Meaning
Doctor_Name	Doctor name
Specialization	Medical specialization
Experience	Doctor experience
Hospital_Name	Hospital/clinic name
City	Practice city
City_Tier	Metro or Non-Metro classification
Patient_Rating(%)	Patient recommendation/rating percentage
Source	Data source
Consultation_Fee(₹)	Consultation fee in Indian Rupees

6. Data Cleaning & Analytical Preparation

The cleaning stage separates data-quality corrections from analytical preparation. This preserves a clean checkpoint before derived analytical features are introduced.

- Patient-rating values were converted from percentage text to numeric values.
- Missing patient-rating values were imputed using the median.
- "Free Consultation" entries were converted to ₹0 to preserve their economic meaning.
- Consultation fees were converted to numeric format.
- Hospital names were standardized where hyphen-separated text created inconsistent naming.
- Duplicate records were checked; no duplicates were found.
- Consultation-fee outliers were identified using the IQR method but retained in the canonical dataset.
- Six cities were classified as Metro and the remaining six as Non-Metro through the City_Tier feature.

6.1 Outlier Strategy

IQR Metric	Value
Q1	₹500
Q3	₹1,000
IQR	₹500
Lower bound	-₹250
Upper bound	₹1,750
Outliers above upper bound	39

The 39 fee outliers were retained because an unusually high consultation fee may be a legitimate market observation rather than a data error. An outlier-excluded view was used only as a sensitivity analysis to understand how extreme observations affect the overall fee distribution.

7. Exploratory Data Analysis

7.1 Overall Fee Landscape

The average consultation fee is ₹829.35, while the median is ₹800. The range spans ₹0 to ₹4,000, with 27 free-consultation records. The mean being slightly above the median is consistent with a right-skewed distribution.

Measure	Value
Mean	₹829.35
Median	₹800
Minimum	₹0
Maximum	₹4,000
Standard deviation	₹434.86
Free consultations	27

The outlier-excluded sensitivity view reduces the average fee to approximately ₹779.54 and the median to ₹750, demonstrating the effect of the high-fee tail.

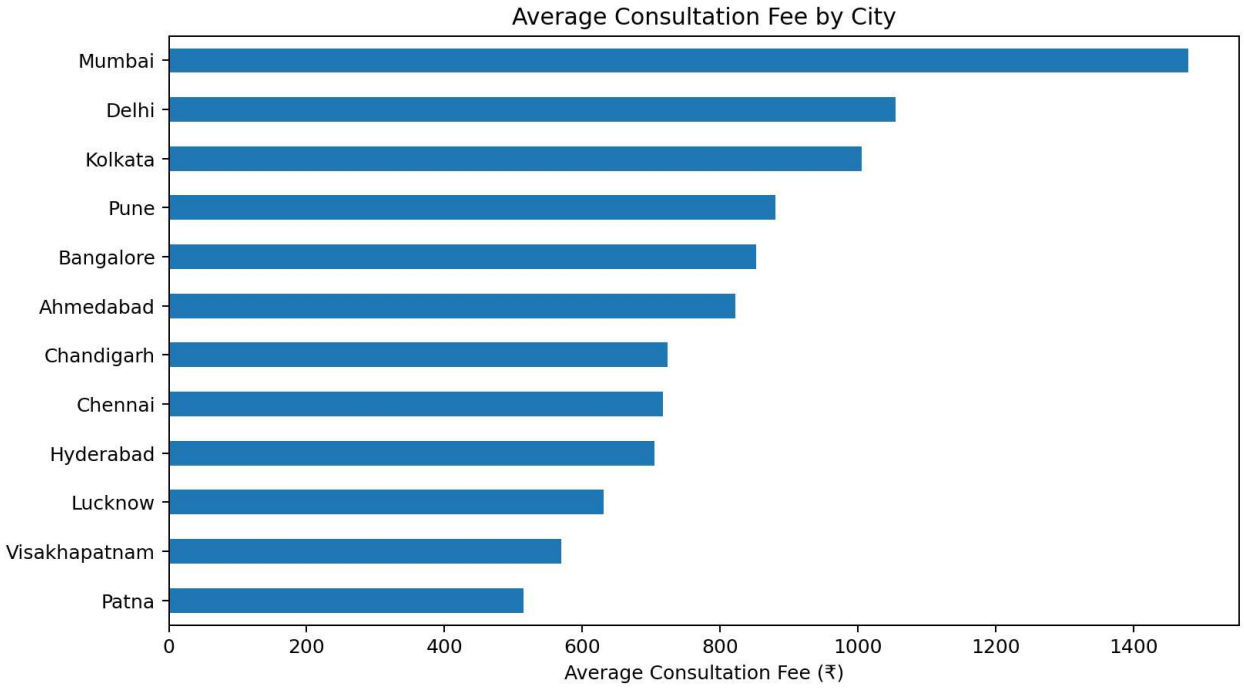


Figure 1. Average consultation fee by city.

7.2 City-Wise Pricing

City-level analysis reveals substantial variation in observed consultation pricing. Mumbai has the highest average consultation fee, while Patna has the lowest.

Rank	City	Average Fee
1	Mumbai	₹1,478.89
2	Delhi	₹1,054.44
3	Kolkata	₹1,005.00
4	Pune	₹879.42
5	Bangalore	₹852.22
6	Ahmedabad	₹822.22
—	Patna (lowest)	₹515.00

The city comparison is descriptive: it identifies where observed pricing is higher or lower within this dataset, rather than establishing why those differences occur.

7.3 Metro vs Non-Metro

Metro cities have an average consultation fee of ₹968.70, compared with ₹690.00 in Non-Metro cities. The observed average difference is ₹278.71.

City Tier	Average Fee	Median Fee	Records
Metro	₹968.70	₹900	540
Non-Metro	₹690.00	₹700	540

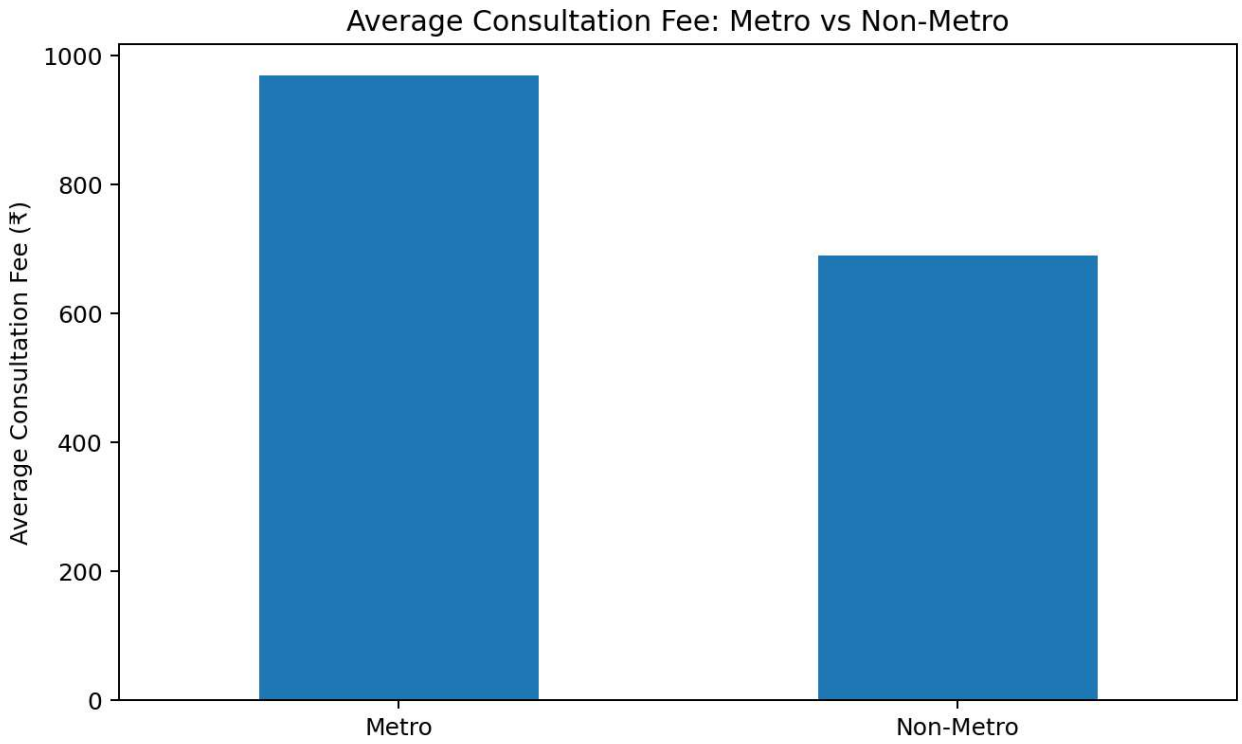


Figure 2. Average consultation fee by city tier.

7.4 Specialization-Wise Pricing

Specialization is another strong descriptive pricing dimension. Cardiologists have the highest average fee, while Dentists have the lowest.

Rank	Specialization	Average Fee
1	Cardiologist	₹1,016.25
2	Neurologist	₹991.25
3	General Surgeon	₹866.67
4	Orthopedist	₹859.58
5	General Physician	₹858.75
6	Dermatologist	₹839.58
7	ENT	₹814.17
8	Gynecologist	₹792.50
9	Dentist	₹425.40

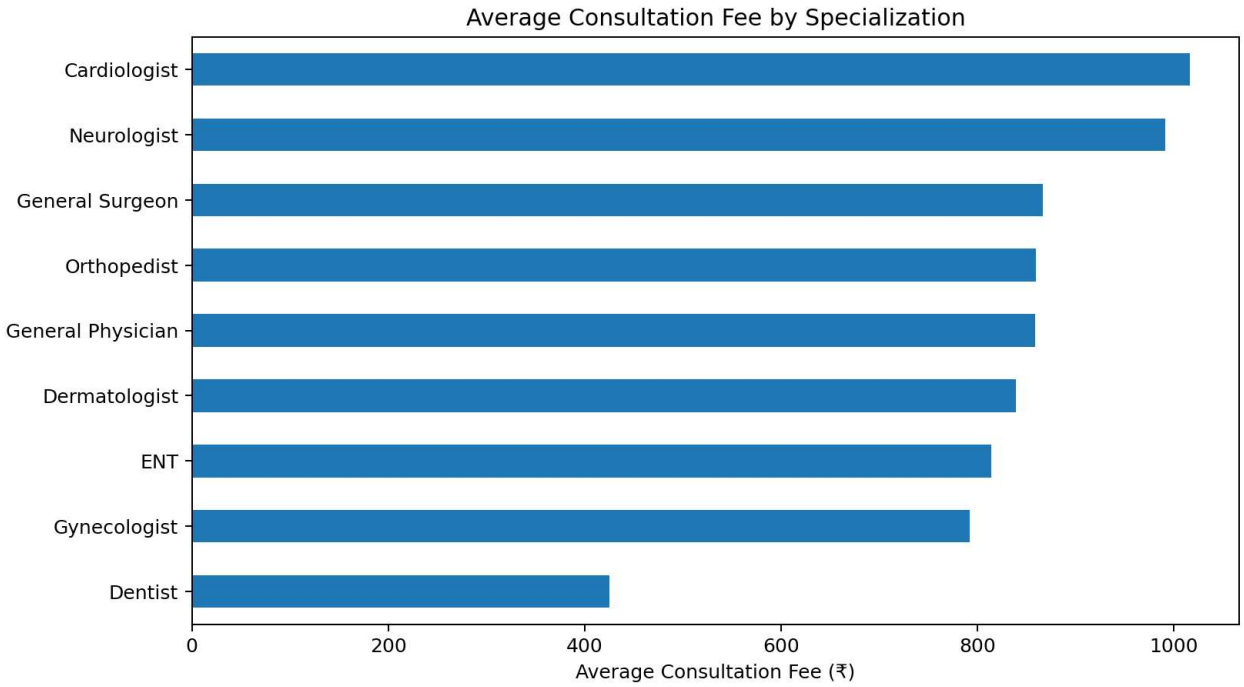


Figure 3. Average consultation fee by medical specialization.

7.5 Fee Variability

Specialization	Variance	Std. Dev.	Fee Range
General Surgeon	247,661	₹497.66	₹4,000
Neurologist	238,053	₹487.91	₹2,500
Orthopedist	200,475	₹447.74	₹2,600
Cardiologist	196,856	₹443.68	₹2,500
ENT	172,445	₹415.26	₹3,000
General Physician	156,624	₹395.76	₹2,500
Gynecologist	117,128	₹342.24	₹2,000
Dermatologist	111,424	₹333.80	₹2,000
Dentist	42,865	₹207.04	₹1,500

General Surgeon shows the highest fee variance, indicating greater dispersion within that specialization. Dentists showed the lowest variance, indicating comparatively more consistent pricing. Variability itself does not explain its underlying cause.

7.6 Experience vs Consultation Fee

The Pearson correlation between experience and consultation fee is 0.142, indicating a weak positive linear association. The broad spread of fees at similar experience levels shows that experience alone does not explain consultation pricing.

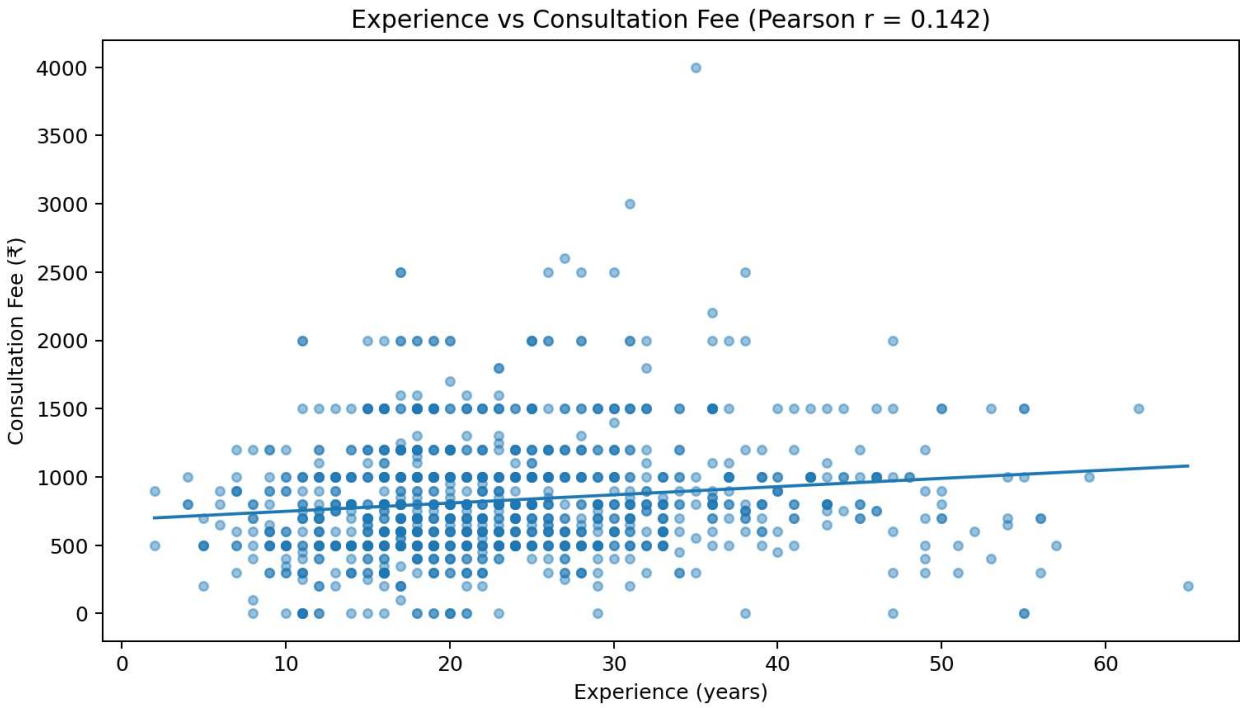


Figure 4. Experience and consultation fee relationship.

7.7 Patient Rating vs Consultation Fee

The Pearson correlation between patient rating and consultation fee is -0.133, indicating a weak negative linear association. Higher patient ratings do not necessarily correspond to higher consultation fees.

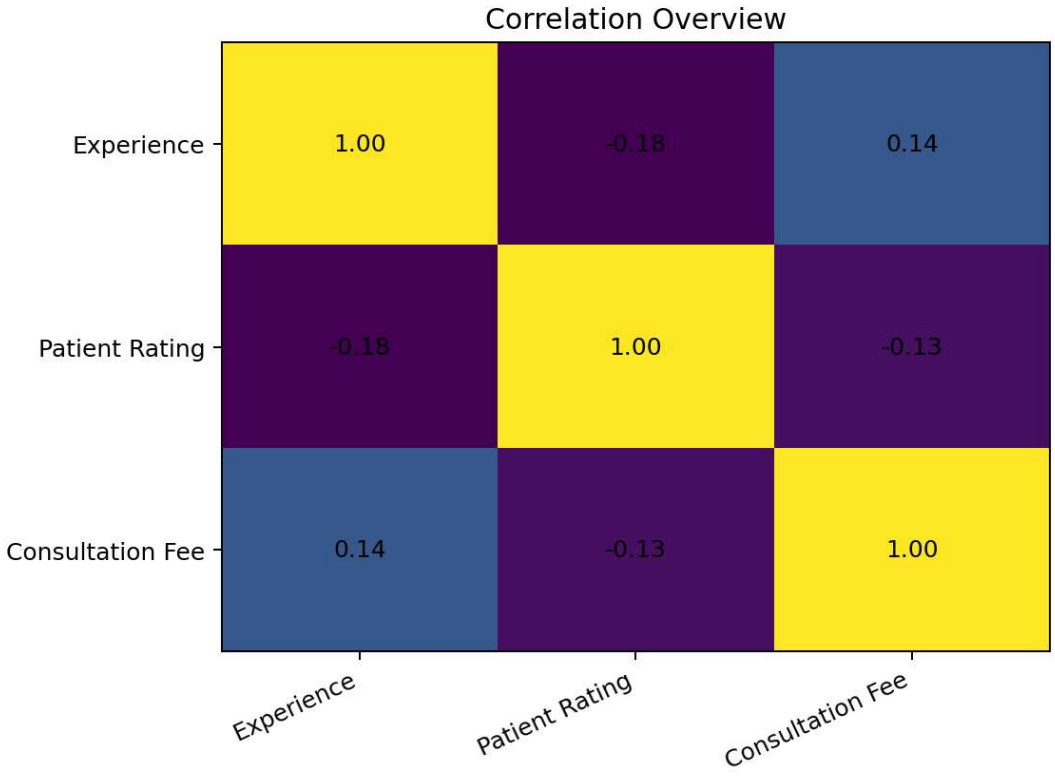


Figure 5. Correlation overview for experience, patient rating, and consultation fee.

7.8 City × Specialization Pricing

The city–specialization analysis shows that the same specialization can have substantially different observed consultation fees across cities. This highlights localized pricing differences and makes the combination of geography and medical service useful for comparative analysis.

8. Premium Pricing Analysis

The premium group is defined using the 75th percentile of the canonical consultation-fee distribution. The threshold is ₹1,000; 365 records (33.8%) fall at or above this threshold.

Premium Metric	Result
Threshold	₹1,000
Premium records	365
Share of records	33.8%
Highest premium share by city	Mumbai (87%)
Highest premium share by specialization	Neurologist (53%)

Because repeated fee values can occur at the percentile threshold, the resulting premium share can be slightly above 25%. The premium category is therefore a threshold-based descriptive segment, not a claim that exactly one quarter of doctors are premium.

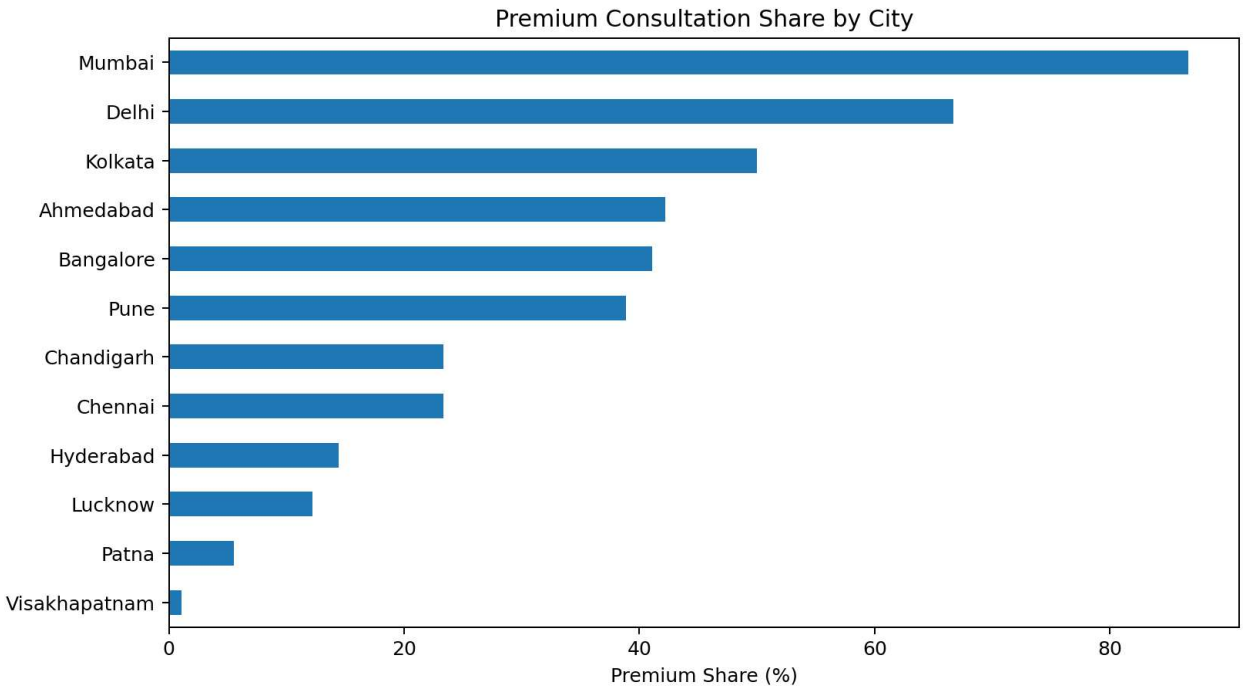


Figure 6. Share of doctors at or above the premium fee threshold by city.

Premium Concentration

- Mumbai has the highest premium consultation share among cities.
- Neurology has the highest premium consultation share among specializations.
- Premium pricing varies across city–specialization combinations, indicating localized high-fee clusters.

9. Fee Segmentation

For relative market segmentation, the analysis uses paid consultations only so that free consultation records do not automatically form the lowest-priced group.

Segment	Doctors	Share
Budget	366	34.8%
Mid-range	467	44.3%
Premium	220	20.9%

These segments are relative to the paid-fee distribution in this dataset. They are not universal rupee-based healthcare market categories.

10. Business Insights

Geographic pricing differentiation: Metro markets show higher observed consultation pricing than Non-Metro markets, and Mumbai stands out as the highest-priced city in the dataset.

Specialization-based pricing: Consultation fees vary materially across medical specializations, with Cardiologists and Neurologists at the higher end and Dentists at the lower end.

Premium market concentration: Higher-priced consultations are concentrated in specific cities and specializations rather than being evenly distributed across the market.

Limited explanatory power of experience and ratings: Experience and patient ratings show weak linear relationships with fees, so neither variable alone provides a strong explanation of consultation pricing.

Localized pricing opportunities: City × specialization analysis reveals that pricing for the same medical service can differ meaningfully by location, which is useful for benchmarking and market comparisons.

How this can support business decisions

The analysis can support comparative price benchmarking, market-level positioning, specialization-level pricing reviews, and identification of higher-priced city–service combinations. These are analytical use cases suggested by the observed data, not forecasts or causal conclusions.

11. Analytical Recommendations

13. Use city and specialization as primary segmentation dimensions when benchmarking consultation fees.
14. Treat experience and patient rating as supplementary context rather than standalone pricing predictors.
15. Use premium-fee concentration to identify markets that warrant deeper pricing investigation.
16. Retain outliers in the primary dataset while using sensitivity analysis to assess their effect.

17. For future analysis, add richer explanatory variables such as doctor credentials, clinic type, consultation mode, location-level socioeconomic indicators, and time/date information if available.

12. Limitations & Scope

- The dataset is observational and describes the collected Practo records; it does not establish causal relationships.
- The analysis represents the project dataset and should not be interpreted as a complete or current census of Indian healthcare pricing.
- The premium threshold is dataset-relative and should not be treated as an industry-wide definition of premium consultation pricing.
- Patient rating is used as an observed rating/recommendation measure; it is not a direct measure of service quality or reputation.
- The current Practo website may change over time. The project therefore preserves the historical 1,080-record dataset rather than assuming a fresh live scrape will reproduce it.

13. Reproducibility & Project Assets

The repository is organized as a staged analytical workflow. The notebooks are intended to be read and executed in sequence, while the processed datasets preserve checkpoints between cleaning and analysis.

Asset	Purpose
data/raw/practo_raw_data.csv	Raw project dataset before cleaning
data/processed/practo_cleaned_data.csv	Cleaned dataset after data-quality corrections
data/processed/practo_analysis_data.csv	Analysis-ready dataset with standardized fields and City_Tier
notebooks/01_data_extraction.ipynb	Extraction methodology and controlled live scraping demonstration
notebooks/02_data_cleaning.ipynb	Cleaning, validation, and analytical preparation
notebooks/03_exploratory_data_analysis.ipynb	EDA, comparisons, correlations, premium analysis, and segmentation
presentation/doctor_consultation_fee_analysis.pptx	Presentation of project findings
requirements.txt	Python dependencies
README.md	Project documentation and usage guide

14. Conclusion

The Doctor's Consultation Fee Analysis project demonstrates a complete exploratory analytics workflow—from web-sourced doctor-level data through cleaning, analytical preparation, and business-oriented EDA.

The strongest observed pricing differences are associated with geography and medical specialization. Mumbai records the highest average consultation fee, Metro markets average more than Non-Metro markets, and Cardiologists have the highest specialization-level average fee. In contrast, experience and patient rating show only weak linear associations with consultation fees.

The project also demonstrates responsible analytical practice by retaining potentially legitimate fee outliers in the canonical dataset, using sensitivity analysis rather than automatically deleting high-value observations, and clearly separating observed associations from causal explanations.

Final perspective
The project is best viewed as a descriptive pricing-benchmarking analysis: it identifies where consultation fees differ and where premium pricing is concentrated, creating a foundation for deeper market research or predictive analysis with richer explanatory data.

Appendix A — Selected City Benchmarks

City	Average Fee	Premium Share
Mumbai	₹1,478.89	87%
Delhi	₹1,054.44	67%
Kolkata	₹1,005.00	50%
Pune	₹879.42	39%
Bangalore	₹852.22	41%
Ahmedabad	₹822.22	42%
Chandigarh	₹723.89	23%
Chennai	₹716.67	23%
Hyderabad	₹705.00	14%
Lucknow	₹630.56	12%
Visakhapatnam	₹568.89	1%
Patna	₹515.00	6%

Appendix B — Selected Specialization Benchmarks

Specialization	Average Fee	Premium Share
Cardiologist	₹1,016.25	50%
Neurologist	₹991.25	53%
General Surgeon	₹866.67	36%
Orthopedist	₹859.58	30%
General Physician	₹858.75	36%
Dermatologist	₹839.58	31%
ENT	₹814.17	33%
Gynecologist	₹792.50	32%
Dentist	₹425.40	3%